MIL-M-0038510/23C(USAF)
1 December 1975
USED IN LIEU OF
MIL-M-38510/23A
14 July 1973
SUPERSEDING
MIL-M-0038510/23B(USAF)
15 October 1973

MILITARY SPECIFICATION

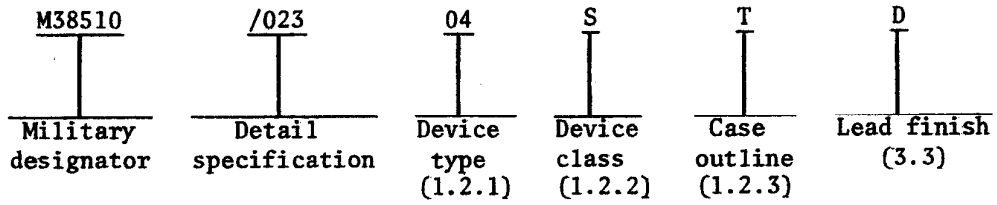
# MICROCIRCUITS, DIGITAL, TTL, HIGH SPEED, NAND GATES, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/23A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

## 1. SCOPE

1.o Scope. This specification covers the requirements for monolithic silicon, TTL, high speed, positive NAND gate microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:




1.2.1 Device type. The device type shall be as shown in the following:

| Device type | Circuit | Similar commercial type |
|---|---|---|
| 04 | Quadruple, 2-Input NAND Gate | 54H00 |
| 05 | Hex, 1-input inverter gate | 54H04 |

1.2.2 Device class. The device class shall be class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range - - - - - - - - - | -0.5 Vdc to 7.0 Vdc |
| Input voltage range - - - - - - - - - - | 0 to 5.5 Vdc |
| Storage temperature range - - - - - | -65°C to 150°C |
| Lead temperature (soldering, 10 seconds) - - - - - - - - - - - - - | 300°C |

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1.4 Recommended operating conditions.

| | |
|---|---|
| Ambient operating temperature range | -55°C to 125°C |
| Supply voltage | 4.5 Vdc minimum to 5.5 Vdc maximum |
| Minimum high level input voltage | 2.0 Vdc |
| Maximum low level input voltage | 0.8 Vdc |

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids request for proposal, forms a part of this specification to the extent specified herei

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordanc with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, constructi weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth table and logic equations. The truth table and logic equations shall as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordanc with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance character tics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 an as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall con sist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.



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TABLE I. Electrical characteristics tests (Device type 04).

| Test No. 1/2/ | Characteristics | Symbol | Conditions | Limits 3/ | | Delta (Δ) 1/ | Units |
|---|---|---|---|---|---|---|---|
| | | | | Min | Max | | |
| 1 | Input voltage required to ensure a high level input (each input) | $V_{IH}$ | $V_{CC}$ = 4.5 V | 2 | | | V |
| 2 | Input voltage required to ensure a low level input (each input) | $V_{IL}$ | $V_{CC}$ = 4.5 V | | 0.8 | | V |
| 3* | High level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V, $V_{IH}$ = 2 V, $V_{IL}$ = 0.8 V, $I_{LOAD}$ = -500 µA | 2.4 | | ±.12 | V |
| 4* | Low level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V, $V_{IH}$ = 2 V, $V_{IL}$ = 0.8 V, $I_{SINK}$ = 20 mA | | 0.4 | ±.04 | V |
| 5* | High level input current (each input) | $I_{IH}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V | | 50 | 10X or 5 µA | µA |
| | | | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V | | 100 | N/A | |
| 6 | Low level input current (each input) | $I_{IL}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | -1.0 | -2.0 | | mA |
| 7 | Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V, $V_{IH}$ = 4.5 V, $V_{IL}$ = 0 V | -40 | -100 | | mA |
| 8 | Supply current | $I_{CCL}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 4.5 V | | 40 | | mA |
| 9 | Supply current | $I_{CCH}$ | $V_{CC}$ = 5.5 V, $V_{IH}$ = 4.5 V, $V_{IL}$ = 0 V | | 16.8 | | mA |
| 10 | Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V, $I_{IC}$ = -8 mA, $T_A$ = 25°C | | -1.5 | | V |

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at $T_A$ = -55°C to +125°C.

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TABLE I. Electrical characteristics tests (Device type 05).

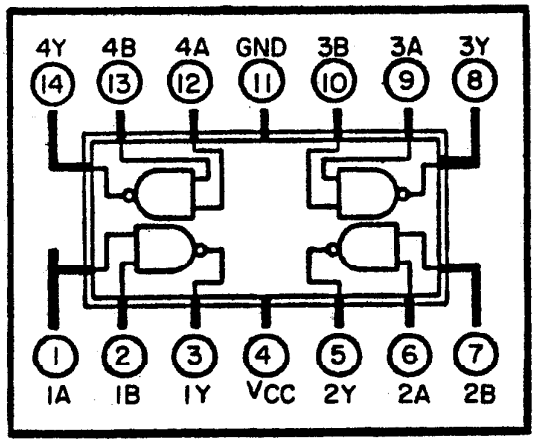
| Test No. 1/2/ | Characteristics | Symbol | Conditions | Limits 3/ | | Delta (Δ) 1/ | Units |
|---|---|---|---|---|---|---|---|
| | | | | Min | Max | | |
| 1 | Input voltage required to ensure a high level input (each input) | $V_{IH}$ | $V_{CC}$ = 4.5 V | 2 | | | V |
| 2 | Input voltage required to ensure a low level input (each input) | $V_{IL}$ | $V_{CC}$ = 4.5 V | | 0.8 | | V |
| 3* | High level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V, $V_{IH}$ = 2 V<br>$V_{IL}$ = 0.8 V, $I_{OH}$ = -500 µA | 2.4 | | ±.12 | V |
| 4* | Low level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V, $V_{IH}$ = 2 V<br>$V_{IL}$ = 0.8 V, $I_{OL}$ = 20 mA | | 0.4 | ±.04 | V |
| 5 | Low level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | -1 | -2 | | mA |
| 6* | High level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V | | 50 | ±5.0 | µA |
| 7 | High level input current | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V | | 100 | | µA |
| 8 | Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0 V | -40 | -100 | | mA |
| 9 | Supply current | $I_{CCL}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V | | 60 | | mA |
| 10 | Supply current | $I_{CCH}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0 V | | 26 | | mA |
| 11 | Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V, $I_{IN}$ = -8 mA<br>$T_A$ = 25°C | | -1.5 | | V |

1/ Delta computations shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at $T_A$ = -55°C to +125°C.



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Device type 04




Device type 05

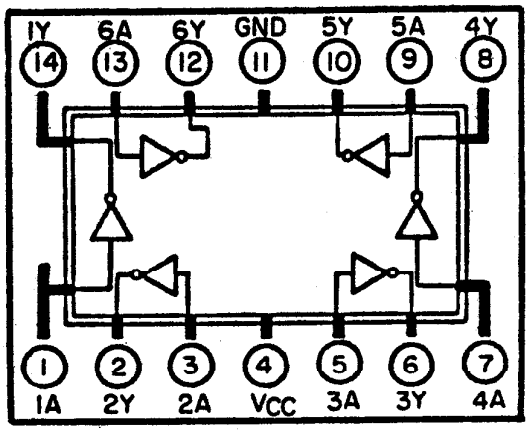



FIGURE 1. Logic diagram and terminal connections.

Device type 04

| Truth table each gate | | |
|---|---|---|
| Input | | Output |
| A | B | Y |
| L | L | H |
| H | L | H |
| L | H | H |
| H | H | L |

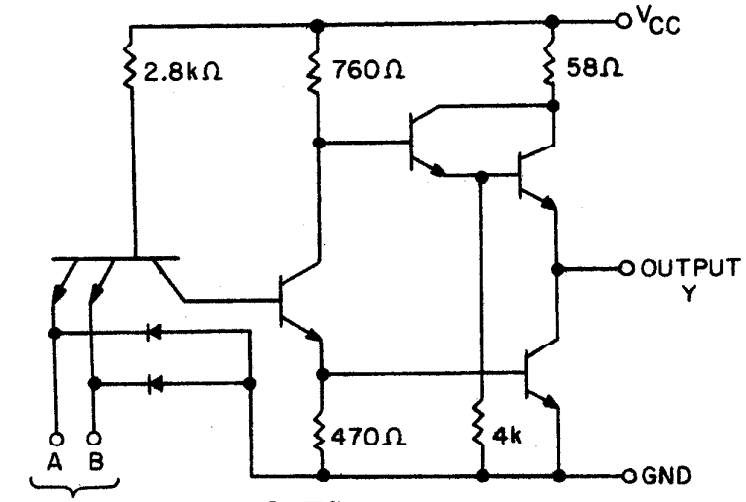
Positive logic $Y = \overline{AB}$

Device type 05

| Truth table each gate | |
|---|---|
| Input | Output |
| A | Y |
| L | H |
| H | L |

Positive logic $Y = \overline{A}$

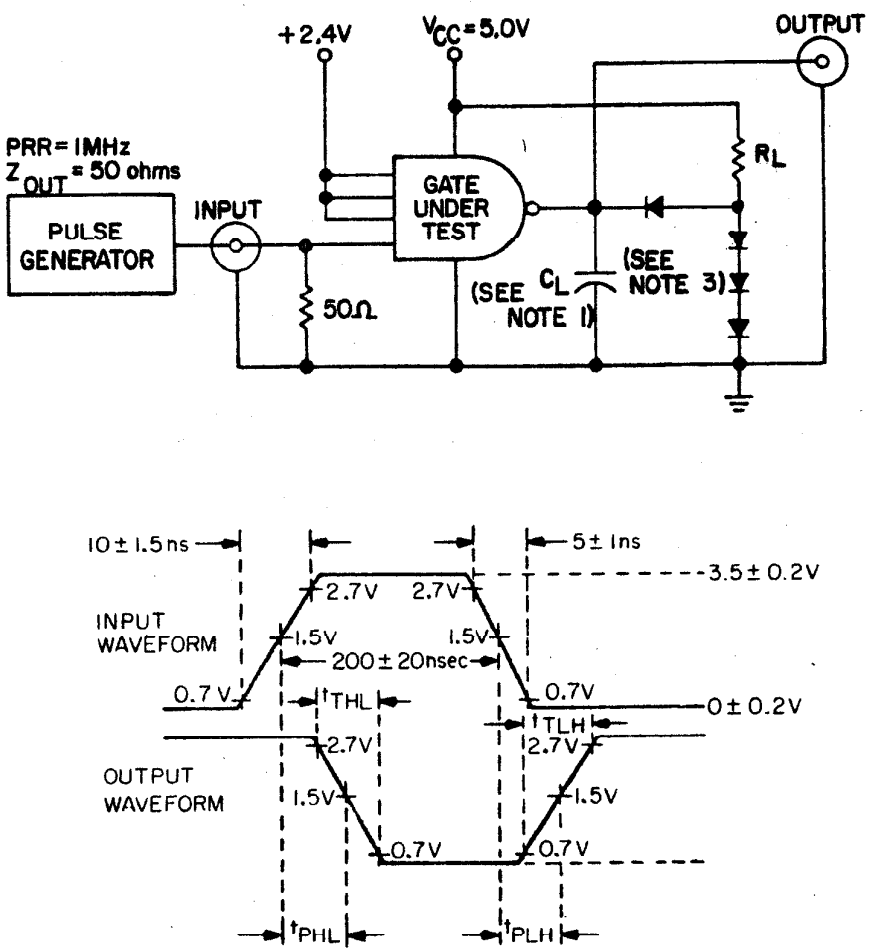
FIGURE 2. Truth tables and logic equations.




NOTES:
1. Device type 05 has one input.
2. Component values shown are nominal.

FIGURE 3. Schematic circuit.

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NOTES:
1. $C_L$ = 50 pF minimum, including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground termina
3. All diodes are 1N3064 or equivalent.
4. $R_L$ = 280Ω ±5%.

FIGURE 4. Switching time test circuit.



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3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

TABLE II. Switching characteristics (Device type 04).

| Characteristic 1/ | Symbol | Conditions | Test figure | Temp. | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| Propagation delay time, high-to-low level | $t_{PHL}$ | $V_{CC}$ = 5.0V<br>$C_L$ = 50 pF<br>$R_L$ = 280Ω | 4 | 25°C | 2 | 12 | ns |
| | | | | 125°C | 2 | 14 | |
| | | | | -55°C | 2 | 16 | |
| Propagation delay time, low-to-high level | $t_{PLH}$ | | | 25°C | 2 | 12 | ns |
| | | | | 125°C | 2 | 15 | |
| | | | | -55°C | 2 | 13 | |

1/ Test each gate.

TABLE II. Switching characteristics (Device type 05).

| Characteristic 1/ | Symbol | Conditions | Test figure | Temp. | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| Propagation delay time, high-to-low level | $t_{PHL}$ | $C_L$ = 50 pF<br>$R_L$ = 280 Ω ± 5%<br>$V_{CC}$ = 5.0 V | 4 | 25°C | 2 | 12 | ns |
| | | | | -55°C<br>125°C | 2 | 16 | ns |
| Propagation delay time, low-to-high level | $t_{PLH}$ | | 4 | 25°C | 2 | 12 | ns |
| | | | | -55°C<br>125°C | 2 | 15 | ns |

1/ Test each gate.

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3.6 Extended life tests. Extended life tests shall be performed in accordance wi MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-



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